Midterm Practical Exam

Exam Guidelines

- Time limit: 60 minutes. Programming language: Java.
- Create an Apache NetBeans Java Application project named YourLastname+PracticalExamC (example: EspinaPracticalExamC). Create PracticalExamC.java and ClientHandler.java in the same package under Source Packages.
- Work individually using Apache NetBeans IDE only. Do not use other IDEs, text editors, browsers, external terminals, or AI tools. Write, build, and run your program inside NetBeans.
- Use port 12345 unless the instructor assigns a different available port. Build without errors and keep the server running in NetBeans for the instructor's Telnet test.
- Submit the complete NetBeans project folder, including both Java source files and the project configuration, after demonstrating the program.

Problem

Lowercase Message Server. A computer laboratory needs a server that converts messages to small letters. Create a simple multithreaded TCP server that handles two clients at the same time and repeatedly responds to each client's messages.

Required Specifications

1. Create PracticalExamC as the main class. Use ServerSocket to listen on the required port and print "Server is listening on port 12345" (or the assigned port).

2. Keep accepting connections in a loop. For every accepted Socket, start a new Thread with a ClientHandler that implements Runnable. Keep client reading and writing inside the handler.

3. In ClientHandler, read one line at a time using BufferedReader and InputStreamReader. Send each reply using PrintWriter with automatic flushing enabled.

4. For every input other than exit, return Server: followed by one space and the message converted to lowercase. You may use String.toLowerCase().

5. Continue reading and replying on the same connection until the client sends exit (ignore letter case) or disconnects. For exit, reply Goodbye! and close only that client's connection.

6. Keep the server and other clients running when one client leaves. Close the client socket and streams properly and handle IOException. A graphical interface and a Java client program are not required.

Apache NetBeans Setup Build and Run

1. Choose File > New Project > Java with Ant > Java Application. Enter your required project name. Clear Create Main Class, then click Finish.

2. Under Source Packages, use New > Java Class to create PracticalExamC and ClientHandler in the package practicalexam. Keep the same package declaration in both files.

3. Write the required code and save all files. Right-click the project and choose Clean and Build. Correct any errors shown in the Output window.

4. Right-click the project and choose Run. If prompted to select a main class, choose practicalexam.PracticalExamC. Check the listening message in the Output window and leave the server running.

Expected Responses

The instructor sends these inputs through Telnet. Your server must return the matching responses.

Client input	Server response
HELLO CLASS	Server: hello class
Java Is Fun	Server: java is fun
TEST 123	Server: test 123
exit	Goodbye!

Instructor Test

Only the instructor operates Telnet. The instructor connects two clients to your server's IP address and assigned port, sends at least two messages from each, and checks that Client 2 receives replies while Client 1 stays connected and idle.

The instructor sends exit from Client 1 and confirms that Client 2 still receives replies and a new client can connect. EXIT must also return Goodbye! Students remain in NetBeans throughout the test.

"Iteratively" means reading and responding repeatedly within each client's handler. The server must still handle multiple clients concurrently.

Evaluation Criteria

Total: 100%

Criteria	Percentage
Server Setup & Functionality (listening & accepting)	30%
Handling Instructor's Requests Iteratively	25%
Correct Implementation of Java I/O Streams	15%
Successful Response to Instructor's Telnet Test	15%
Code Clarity	10%
Compliance with Exam Rules (no errors, proper port usage, no AI/other tabs)	5%
Total	100%